Check for updates

Corporate Social Responsibility and Environmental Management

WILEY

Corporate Social Responsibility and Environmental Management

RESEARCH ARTICLE OPEN ACCESS

Greenwashing in Brazilian Corporations: A Machine Learning Approach to Unmask the Discourse-Practice Gap

Isabelly Alves Alvares dos Santos | Ana Cláudia de Araújo Moxotó | Edjard de Souza Mota

Federal University of Amazonas, Manaus, Brazil

Correspondence: Ana Cláudia de Araújo Moxotó (anaclaudiamoxoto@ufam.edu.br)

Received: 6 June 2025 | **Revised:** 1 August 2025 | **Accepted:** 14 August 2025

Funding: This work was supported by Coordenação de Aperfeiçoamento de Pessoal de Nível Superior.

Keywords: Artificial Intelligence | corporate social responsibility | decoupling theory | emerging markets | ESG | greenwashing

ABSTRACT

This study investigates greenwashing in the Brazilian corporate sector by analysing the gap between companies' sustainability claims and their actual practices. We introduce a novel, AI-powered methodology based on Decoupling Theory to detect greenwashing. Our approach triangulates three distinct data sources from 2022 to 2023: (i) internal corporate discourse (ESG reports quantified via machine learning); (ii) external market perception (Merco ESG reputation index); and (iii) actual environmental practice (objective greenhouse gas (GHG) emissions data). Our econometric analysis reveals a significant discourse-practice gap, providing strong evidence of greenwashing. We found a substantial misalignment between companies' AI-calculated ESG scores and their external reputation rankings, with only a small fraction of firms showing consistency. More strikingly, the analysis uncovered a paradoxical positive correlation between environmental discourse and emissions: a 1% increase in a company's AI-calculated environmental score is associated with an approximate 5.24% increase in its GHG emissions. This suggests that environmental disclosures may serve as a tool for image management rather than reflecting tangible emission reductions. These findings empirically validate key facets of greenwashing theory within an emerging market context. The results underscore the urgent need for more robust regulation, mandatory standardized reporting, and independent verification to mitigate deceptive environmental claims and foster genuine corporate sustainability.

1 | Introduction

The increasing focus on corporate sustainability has driven a surge in Environmental, Social, and Governance (ESG) disclosures. Yet, this rise is shadowed by greenwashing, a deceptive tactic where companies misrepresent their sustainability efforts to sway stakeholder perceptions (Szabo and Webster 2021; Wang et al. 2023).

While well-documented in developed economies, greenwashing presents a more critical challenge in emerging markets. Here, regulatory systems are often less stringent and information asymmetries more pronounced (Chen and Dagestani 2023). Despite the widespread adoption of ESG reporting among Brazilian firms listed on the B3 stock exchange, there is a significant lack of empirical evidence verifying if these disclosures reflect actual sustainability practices. Prior research, heavily reliant on self-reported data or third-party ESG ratings, is vulnerable to bias and manipulation (Yu et al. 2020; Kalesnik et al. 2020). This absence of robust, data-driven methods for detecting greenwashing in emerging markets obscures the true scope of the problem and erodes stakeholder trust in ESG claims.

In today's environment of increasing pressure for corporate sustainability, organizations face a dilemma: genuinely transform

This is an open access article under the terms of the Creative Commons Attribution License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

© 2025 The Author(s). *Corporate Social Responsibility and Environmental Management* published by ERP Environment and John Wiley & Sons Ltd.

their practices or merely engage in "discursive performativities" that satisfy institutional demands without altering core operations. Decoupling theory, rooted in neo-institutionalism (Meyer and Rowan 1977; Bromley and Powell 2012), provides a vital framework for understanding this phenomenon, explaining how companies construct facades of ESG compliance while their internal practices remain unchanged.

This study investigates the discrepancy between corporate sustainability discourse and actual practice among Brazilian companies to verify the occurrence of greenwashing. To achieve this, it develops and applies a novel methodology powered by Artificial Intelligence (AI) to analyze corporate disclosures. This approach triangulates three distinct data sources: (i) internal discourse, quantified from ESG reports through machine learning (ML); (ii) external perception, measured by Merco ESG index scores; and (iii) actual practice, determined by objective greenhouse gas (GHG) emissions data.

The research addresses gaps in existing literature, which has predominantly focused on developed economies (Seele and Gatti 2017; Delmas and Burbano 2011). Few studies examine emerging markets, where detecting greenwashing can be more challenging (Baldi and Pandimiglio 2022; Zhang 2024). Additionally, while AI and ML have been proposed as tools to detect greenwashing (Zeng et al. 2025; Li et al. 2024), their application in non-English contexts—such as Portuguese-language ESG reports—remains nascent.

This study highlights Brazil as a unique setting to examine greenwashing, contrasting its leadership in renewable energy with persistent environmental controversies (de Freitas Netto et al. 2020). This tension is amplified in natural resource-dependent sectors, where ESG rhetoric often diverges from actual practice. An evolving regulatory landscape further enables selective disclosure (Mateo-Márquez et al. 2022), whereas consumer studies reveal growing skepticism toward eco-claims (Caldas et al. 2021).

Empirical evidence demonstrates greenwashing's sector-wide impacts: eroded consumer trust in retail (Braga Junior et al. 2019), performative sustainability in mega-events (Boykoff and Mascarenhas 2016), and agribusiness disinformation campaigns (Medeiros et al. 2024). Although corporate governance mechanisms show mitigating potential (Duarte et al. 2024), institutional gaps sustain this decoupling between discourse and practice, positioning Brazil as a critical case for understanding greenwashing in emerging economies.

The paper proceeds as follows: Section 2 reviews the literature on ESG reporting, greenwashing, and AI applications in sustainability. Section 3 details the methodology, including ML techniques and econometric models. Sections 4 and 5 present the results and discussion, respectively, whereas Section 6 concludes with policy implications and future research directions. This study advances the field by providing a replicable framework to detect greenwashing in emerging markets, highlighting the urgent need for standardized ESG auditing and AI-driven transparency tools to align corporate discourse with genuine sustainability action.

2 | Literature Review and Hypothesis

2.1 | Decoupling Theory and Greenwashing

Decoupling theory provides a framework for understanding the gap between organizational rhetoric and action. Originally developed by Meyer and Rowan (1977), it explains how companies adopt sustainability programs as "rational myths" to gain legitimacy rather than to drive real change. Bromley and Powell (2012) expanded this view, showing how organizations focus on creating monitoring systems (means) while losing sight of actual environmental goals (ends), resulting in perpetual reform without progress. This theoretical foundation helps explain greenwashing, which Delmas and Burbano (2011) define as a form of decoupling where companies create a false impression of environmental responsibility. Recent research reveals how this phenomenon operates in practice: Bothello et al. (2023) found corporate structures enable decoupling, whereas Seele and Gatti (2017) identified "machinewashing" through algorithmic manipulation. Empirical studies show 7% of companies engage in greenwashing, particularly in extractive industries (Keresztúri et al. 2025), with traditional ESG ratings proving vulnerable to manipulation (Yu et al. 2020).

The Brazilian context illustrates these dynamics clearly. Companies across sectors exhibit significant gaps between sustainability claims and actions. While some firms like Petrobras and Vale prominently feature ESG terms in reports, they remain among the country's largest polluters. This pattern mirrors findings from Braga Junior et al. (2019) on consumer deception and Boykoff and Mascarenhas' (2016) analysis of the Rio Olympics' unfulfilled environmental promises. Even as governance mechanisms show potential to reduce greenwashing (Duarte et al. 2024), weak regulation and information asymmetry allow the practice to persist. The result is a systemic disconnect between corporate sustainability discourse and environmental performance, with significant consequences for stakeholders and regulatory systems alike.

2.2 | ESG Reporting and the "Talk Versus Walk" Discrepancy

ESG reports serve as vital instruments for communicating non-financial performance, documenting corporate commitments to environmental stewardship, social responsibility, and governance integrity (Mishra and Schmidt 2018). The growing imperative of climate action and stakeholder demands has accelerated the adoption of standardized ESG frameworks like the Global Reporting Initiative (GRI) and Corporate Sustainability Assessment (CSA), which aim to enhance transparency and comparability across organizations (Moodaley and Telukdarie 2023; Mougenot and Doussoulin 2022). However, this push for disclosure has created opportunities for greenwashing, particularly among financially constrained firms operating in liberalized capital markets, while state-owned enterprises and those subject to mandatory reporting requirements demonstrate more reliable disclosures (Liu et al. 2024; Brito et al. 2022). The fundamental challenge lies in the discrepancy between corporate sustainability rhetoric and actual practice—what Pizzetti et al. (2021) term the divide between "responsible words" and "irresponsible walks." When companies

15353966, 2025, 6, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/csr.70133, Wiley Online Library on [18/06/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

engage in such deceptive reporting, they not only undermine investor decision-making (Yu et al. 2020) but also erode consumer trust (Seele and Gatti 2017) and employee engagement. Research demonstrates that greenwashing fosters organizational cynicism and diminishes pro-environmental behaviors among workers by creating perceptions of corporate hypocrisy (Tao and Chao 2024). Conversely, authentic ESG implementation can yield competitive advantages, as evidenced by Wu et al. (2020)'s findings that transparent firms attract greater investment and market differentiation.

2.3 | ESG, Greenwashing and Carbon Emissions

The rise of environmental awareness has paradoxically led to increased greenwashing, where companies exaggerate sustainability efforts while concealing negative impacts (de Freitas Netto et al. 2020; Lyon and Maxwell 2011). This practice distorts investment decisions by creating a disparity between reported ESG performance and actual environmental impact (Yu et al. 2020). Beyond misleading stakeholders, greenwashing undermines corporate value, stifles innovation, and erodes trust in sustainable development initiatives (Ghitti et al. 2023; Wang et al. 2023). Its effects are particularly damaging in the financial sector, where it disrupts resource allocation and systemic stability (Baldi and Pandimiglio 2022).

Current regulatory frameworks remain inadequate to address these challenges. The lack of standardized ESG reporting and independent verification enables selective disclosure, especially among multinational corporations (Toscano et al. 2022). Effective solutions require multi-stakeholder collaboration, combining government regulation, media oversight, and independent auditing to validate ESG claims (Wang et al. 2023; Delmas and Burbano 2011). Recent innovations like green subsidies and labeling have shown promise in curbing deceptive practices (Gao and Wei 2025).

This study examines these issues through the lens of "carbon-washing," where companies maintain high emissions despite positive ESG ratings (In and Schumacher 2021; Treepongkaruna et al. 2024). Our analysis tests whether corporate ESG discourse aligns with external validation and actual emissions data (H1). The findings reveal a troubling pattern: many firms with strong ESG communications show poor environmental performance, suggesting that current reporting often serves as image management rather than reflecting substantive action.

The regulatory landscape presents both challenges and opportunities. While larger firms respond positively to stricter reporting requirements (Luu et al. 2025), the high costs of decarbonization continue to incentivize short-term profit maximization in carbon-intensive industries (Senna and de Araujo Moxotó 2025). Market mechanisms alone appear insufficient—Atilgan et al. (2024) demonstrate that emission-intensive firms often enjoy higher returns, underscoring the need for robust policy interventions to align financial incentives with environmental goals.

To directly investigate the discrepancy between discourse and practice, this hypothesis examines the link between a firm's ESG narrative and its tangible environmental impact:

H1. *There is a negative and significant correlation between a firm's ESG discourse score (quantified by AI) and its real-world environmental performance, measured by its greenhouse gas (GHG) emissions.*

This hypothesis tests the discourse-practice gap. A negative correlation is the theoretically expected outcome (i.e., higher discourse corresponds to lower emissions). A failure to confirm this relationship, or the discovery of a positive correlation, would serve as a strong indicator of greenwashing, suggesting that corporate rhetoric is not matched by substantive action.

2.4 | ESG, Greenwashing and AI

Recent studies reveal a paradoxical relationship between ESG performance metrics and corporate sustainability practices. Kathan et al. (2025) demonstrate that elevated ESG scores often correlate with sophisticated greenwashing strategies, especially in large corporations. This finding underscores the critical need for advanced analytical tools to verify sustainability claims. ML emerges as a promising solution, with Zeng et al. (2025) developing optimized algorithms that effectively detect discrepancies in ESG reporting. The growing intersection of AI and sustainability reporting addresses a crucial market need. Moodaley and Telukdarie (2023) highlight how increasing stakeholder demands for environmental accountability have outpaced traditional verification methods. Contemporary AI systems now process vast quantities of sustainability data with unprecedented speed and precision, enabling identification of legitimate green investments (Das et al. 2025) and exposing deceptive reporting practices (Khan et al. 2024). These technological advancements encourage more transparent corporate behavior as organizations recognize the diminishing returns of data manipulation. However, the implementation of AI-based monitoring systems presents its own challenges. Paprocka (2025) emphasizes that without proper governance frameworks, including regulatory oversight and ethical guidelines, even AI solutions risk being co-opted for misleading purposes. The development of standardized audit protocols and transparent algorithmic processes remains essential to ensure these tools fulfill their potential as guardians of corporate sustainability reporting.

3 | Methodology

The primary objective of this research is to investigate the potential occurrence of greenwashing by examining the discrepancy between corporate sustainability discourse and actual practice among Brazilian companies. To achieve this, the study employs a methodology centered on ML techniques to analyze the ESG sustainability reports published by companies.

This approach triangulates three distinct data sources to provide a comprehensive analysis:

- *Internal discourse*: Quantified from corporate ESG reports using ML to create an ESG score.

15353966, 2025, 6, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/csr.70133, Wiley Online Library on [18/06/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

- *External perception*: Measured using the Merco ESG index, which serves as an external reference for evaluating a company's corporate ESG responsibility and reputation.
- *Actual practice*: Determined by objective GHG emissions data, which is compared against the information disclosed in reports.

The sample consists of ESG sustainability reports collected from the websites of one hundred companies from diverse sectors and sizes, all listed on the B3 Brazilian stock exchange. The reports are published annually, and this study uses data from the 2022 and 2023 reference years. The statistical model is based on a cross-sectional data analysis, and the study's robustness is reinforced by conducting linear regressions to confirm the correlations between the different data dimensions.

3.1 | First Phase: ESG Score Measurement—An Approach Using Machine Learning (Internal Discourse)

This study quantifies the content of corporate ESG reports using ML to generate an objective ESG score. This approach is grounded in recent research demonstrating the effectiveness of AI in combating greenwashing. For instance, studies by Lien (2023) and Lagasio (2023) have leveraged ML techniques like "bag-of-words" to process large volumes of data and systematically evaluate greenwashing practices.

Aligned with these methods, this study employs the frequency of a predefined ESG dictionary to calculate an index reflecting the intensity of each company's sustainability discourse. The ML approach helps mitigate subjectivity bias in ESG ratings and introduces a replicable "text as data" method that yields reliable ESG ratings (Mansouri and Momtaz 2022). The method, developed by Mansouri and Momtaz (2022), utilizes a vector language model from large-circulation newspapers to select an ESG corpus. In this study, we used 1498 English reference words based on Mansouri and Momtaz's (2022) implementation. For this research, we developed a Portuguese version of the same corpus (word dictionary), which is in English. We adapted it to the texts of previously collected sustainability reports from Brazilian companies. Therefore, at this initial stage, our method consists of two steps:

a. Build an ESG dictionary in Portuguese based on the English dictionary developed by Mansouri and Momtaz (2022). This dictionary contains ESG sustainability terms extracted from corporate and academic reports. To adapt it to Portuguese, the terms must be translated, their frequency and relevance checked in reliable Portuguese sources, and their consistency and coverage validated.
b. Calculate the (normalised) frequency of ESG terms for each company ("ESG scores"). We use our ESG dictionary to evaluate the E, S, and G dimensions by counting the number of unique occurrences of terms from our dictionary and how many times they occur in sustainability reports issued by companies, normalised by the size of the term set.

While the word frequency counting method is useful for calculating an ESG score, it has a notable limitation: it doesn't consider the context of words, which can lead to incomplete interpretations. To address this and ensure transparency, our study details the construction and validation of a Portuguese ESG dictionary. This involved translating the established English corpus by Mansouri and Momtaz (2022) into Portuguese. Following translation, we conducted a thorough verification of word frequency and relevance using reliable Portuguese sources. Finally, the dictionary's consistency and coverage were validated for the Brazilian context. We acknowledge that the word frequency counting method's inability to fully capture word context is an inherent limitation, but these detailed steps significantly enhance the methodological framework's robustness.

3.2 | Data

The study considered 100 listed companies with the highest market values on the Brazilian stock exchange (B3). Of these, 90 presented sustainability reports, 74 publicly recorded their GHG emissions, and 40 companies were included in Merco ESG. Thus, we arrived at an initial model that includes the dependent and relevant explanatory variables, reported in Table 1.

3.3 | Second Phase: Measuring the Correlation With the Statistical Method

The methodology's second phase statistically evaluates greenwashing potential by correlating AI-quantified ESG discourse with external perception and actual practice. Using OLS regression, the study first examines internal ESG discourse against the Merco ESG index, interpreting a lack of significant positive correlation as greenwashing due to unvalidated sustainability claims. Second, it assesses the link between ESG discourse (aggregate and E, S, G components) and actual GHG emissions via OLS regression. A positive correlation between environmental discourse and actual emissions indicates greenwashing, suggesting misalignment between rhetoric and tangible reduction efforts. Gretl software was used for all statistical analyses.

3.4 | Descriptive Statistics

The statistical analysis reveals distinct patterns across the three key variables. ESG scores demonstrate a symmetrical distribution with mean and median values of 0.201 and a narrow standard deviation (0.0455), indicating consistent reporting practices among most firms. However, scores range from 0.027 (Bradespar SA) to 0.284 (Grupo Neonergia and Itaú), with Cemig (0.281), Banco do Brasil (0.279), and Bradesco (0.273) showing particularly high frequencies of sustainability-related terminology.

Merco ESG reputation scores present a right-skewed distribution (mean 3160 vs. median 4360) with substantial variability (SD = 2890), reflecting wide disparities in corporate reputation. The full scale of scores (0–10,000) suggests significant differences in how companies are perceived relative to their sustainability claims.

15353966, 2025, 6, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/csr.70133, Wiley Online Library on [18/06/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 1 | Description of variables.

Variable name	Acronym	Description	Scale/range	Data source
ESG score	ESG	AI-generated composite score measuring environmental, social and governance performance based on corporate disclosures	0–1 (continuous)	Proprietary AI algorithm analysing sustainability reports
Greenhouse gas emissions	GHG	Total carbon dioxide equivalent emissions reported by the company	Metric tons (tCO_2e)	B3 stock exchange filings and Brazilian National Emissions Registry
Merco ESG responsibility index	Merco	Comprehensive evaluation of corporate ESG performance and reputation	0–10,000 (ordinal)	Merco Corporate Reputation Rankings (https://www.merco.info)

Source: (2023).

Most strikingly, GHG emissions data show extreme positive skewness, with an average (17,542,861.93 tCO_2e) nearly 50 times greater than the median (405,874 tCO_2e). This distribution, characterized by an enormous standard deviation (78,257,287.46), results from extreme outliers like Petrobras Company (588,646,985.5 tCO_2e) at the upper end and Caixa Seguridade Company (211.43 tCO_2e) at the lower bound, highlighting the vast differences in environmental impact across sectors (Table 2).

4 | Estimation Model and Method Result Analysis

If linearity assumptions and correct model specification are met, the estimation approach (statistical regression) OLS (ordinary least squares) or ordinary least squares. However, if hypothesis violations are detected, alternative estimation methods such as the GLS (generalized least squares) estimator, known as generalized least squares or robust errors, must be applied, as they are more efficient in the occurrence of heteroscedasticity.

Gretl was used to carry out linear regression, and a platform was developed to analyse and interpret various data. Gretl is widely used in econometric research to assist in statistical studies. The data is organized cross-sectionally.

i. Model 1:

$$\ln(\text{Merco_ESGi}) = \alpha 0 + \alpha 1 \ln(\text{ESGi}) + \text{ui} \quad (1)$$

The dependent variable Merco_ESG represents the ESG scores of the organizations that the program classifies. Business Corporate—ESG Responsibility (Merco) has been a reference monitor in Latin America since the 2000s and is available online (https://www.Merco.info/br/). For this research, we used data from the last 10 years. The independent variable ESG represents the score the AI method established and is an indicator of occurrences that measure the variation and frequency of words throughout companies' sustainability reports.

ii. Model 2:

$$\ln(\text{GHGi}) = \alpha 0 + \alpha 1 \text{ESGi} + \text{ui} \quad (2)$$

The dependent variable GHG emissions represents the ESG scores of the organizations that the program classifies.

iii. Model 3:

$$\ln(\text{GHGi}) = \alpha 0 + \alpha 1 \ln(\text{Ei}) + \alpha 2 \ln(\text{Si}) + \alpha 3 \ln(\text{Gi}) + \text{ui} \quad (3)$$

The dependent variable GHG emissions represents the E (environment), S (social), G (governance) scores of the organizations that the program classifies.

5 | Result Analysis

5.1 | External Perception

Table 3 presents the results of the regression model considering lnMerco_ESG (presumably the natural logarithm of ESG) as the dependent variable. It also presents the results of the regressions for the independent variable ln_ESG. In general, the results are satisfactory in terms of quality of fit and statistical significance of the coefficients.

The RESET test did not reject the null hypothesis that the Model has an adequate specification and does not present heteroscedasticity, as indicated by the White test. The R^2 is 0.03, indicating that only 3% of the variability of the Merco ESG ranking is explained by the calculated ESG-described variables. The F statistic validates the joint significance of the coefficients. The result of the estimates of the Model used is described in Table 3.

The analysis reveals a concerning disconnect between corporate ESG claims and external validation. Table 3 shows no statistically significant relationship between AI-calculated ESG scores and Merco reputation rankings ($p = 0.6887$), suggesting widespread discrepancies in sustainability reporting. This misalignment indicates potential greenwashing, where companies amplify environmental commitments in disclosures while maintaining unchanged practices. Such strategic exaggeration creates a facade of sustainability that misleads stakeholders and erodes market trust (Delmas and Burbano 2011), particularly when external evaluations fail to corroborate self-reported ESG performance.

15353966, 2025, 6, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/csr.70133, Wiley Online Library on [18/06/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 2 | Statistical description of variables (using observations 1–74).

Variable	Average	Median	Standard deviation	Minimum	Maximum
ESG	0.201	0.201	0.0455	0.027	0.284
Merco	3160	4360	2890	0	10,000
GHG	17,542,861.93	405,874	78,257,287.46	211.43	588,646,985.5

TABLE 3 | Estimation results Merco_ESG.

Variable dependent lnMerco_ESG	Model (1)
constant	8.4683***
p value	< 0.0001
l_ESG	−0.05544
p value	0.6887
R-squared (R^2)	0.03
F-Stat	$F(1, 41)$ 0.16
p value (*F*)	0.68
Heteroscedasticity	LM = 1.84
p value (*F*)	$p(X^2(2) > 1.84) = 0.39$
Specification (RESET test)	$F(2, 43) = 0.24$
p value	$p(F(2, 44) > 0.247503) = 0.78$

Note: Grades: *** indicates that the coefficients are statistically significant at 1% levels, respectively. Standard errors are below the estimates.

5.2 | Actual Practice: GHG Analysis

Table 4 presents a comparative analysis of two OLS regression models designed to explain GHG emissions (represented by ln(GHG)). Model 2 utilizes a single, aggregate (ESG) score as its independent variable, whereas Model 3 disaggregates ESG into its individual environmental (ln(E)), social (ln(S)), and governance (ln(G)) components.

A critical initial step in evaluating these models involves examining their diagnostic tests. The RESET Test for specification revealed a significant difference: Model 3 demonstrated strong evidence of proper specification with a high *p* value of 0.946, indicating its functional form is appropriate. Conversely, Model 2's RESET test yielded a *p* value of 0.057, which, while close to the typical threshold, did not definitively indicate misspecification at the 5% level. Regarding heteroscedasticity, both models performed well, with the White Test yielding *p* values greater than 0.76 for both, confirming the absence of heteroscedasticity and thus supporting the reliability of their standard errors.

Beyond diagnostics, Model 3 consistently outperformed Model 2 in terms of overall fit and explanatory power. Model 3 exhibited a substantially higher R^2 value of 0.329, meaning that its disaggregated E, S, and G components collectively explain 32.9% of the variability in GHG emissions. In contrast, Model 2, using the aggregate ESG score, could only account for 7.8% of this variability. Furthermore, both models demonstrated joint statistical significance through their *F*-tests ($p < 0.05$), rejecting the null hypothesis that all coefficients are zero, with Model 3 showing a notably stronger joint significance ($p = 3.39\text{e-}06$).

This section presents a detailed econometric analysis of Model (2) and Model (3). For Model (2), which examines the impact of a company's ESG score on its GHG emissions (l_GHG), the analysis reveals a striking relationship. Here, the dependent variable, l_GHG, is in logarithmic form, while the independent variable, ESG, is in its original, non-logarithmic scale. It is crucial to note that this ESG score is derived from companies' self-reported sustainability discourse. In contrast, the GHG emissions data are obtained from external sources, such as stock exchanges.

In Model (2), we estimated the impact of a company's non-logarithmic ESG score on its logarithmic GHG emissions (l_GHG). The estimated coefficient for ESG was 17.5508, statistically significant at the 5% level ($p = 0.0157$). This implies that, *ceteris paribus*, a one-unit increase in a company's self-declared ESG score is associated with an approximate 17.55% increase in its GHG emissions. This counterintuitive positive relationship suggests a potential divergence between sustainability rhetoric and tangible environmental performance, possibly indicating greenwashing, or that companies with higher baseline emissions are more active in ESG reporting.

Model (3) further disaggregates this relationship, examining the elasticities of AI-calculated Environmental (log(E)), Social (log(S)), and Governance (log(G)) discourse scores on l_GHG. The Environmental component (log(E)) exhibits a highly significant positive elasticity of 5.24119 ($p < 0.0001$). This indicates that a 1% increase in a company's environmental discourse score is associated with an approximate 5.24% increase in GHG emissions, ceteris paribus. This paradoxical finding suggests that firms with more extensive environmental rhetoric tend to have higher emissions, potentially reflecting industry intensity, a focus on transparency over performance, or outright greenwashing. Conversely, the Social component (log(S)) demonstrates a statistically significant negative elasticity of −2.64926 ($p = 0.0464$). A 1% increase in a company's social discourse score is associated with an approximate 2.65% decrease in GHG emissions, aligning with theoretical expectations that strong social performance can indirectly reduce environmental impact. Similarly, the Governance component (log(G)) shows a marginally significant negative elasticity of −2.35362 ($p = 0.0799$). A 1% increase in governance discourse is associated with an approximate 2.35% decrease in GHG emissions, consistent with robust governance fostering better environmental risk management and emission reduction strategies. While enhancements in social and governance discourse appear to correlate with reduced emissions, increased environmental discourse paradoxically aligns with

15353966, 2025, 6, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/csr.70133, Wiley Online Library on [18/06/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 4 | Estimation results GHG.

Metric/OLS	Model (2) (ESG)	Model (3) (E, S, G components)
Dependent variable	**l_GHG**	**l_GHG**
Constant	9.29601*** (1.45834)	12.9951*** (2.74446)
ESG (aggregate)	17.5508** (7.09117)	—
log(E)	—	5.24119*** (0.901086)
log(S)	—	−2.64926** (1.30626)
log(G)	—	−2.35362* (1.32459)
R^2	0.0784	0.3292
Adjusted R^2	0.0656	0.3004
F-statistic (p value)	6.1257 (0.0157)	11.4509 (3.39e-06)
White Test (p value)	0.7646	0.9575
RESET test (p value)	0.0574	0.9458
Observations	74	74

Note: Grades: ***, **, * indicates that the coefficients are statistically significant at 1% levels of 5% and 10%, respectively. Standard errors are below the estimates.
Source: Compiled by the authors.

higher emissions, underscoring a critical gap between environmental rhetoric and actual performance.

Several factors could explain this: (i) Industry Intensity: Companies in inherently high-emission sectors (e.g., heavy manufacturing, energy) might invest more heavily in environmental disclosure and initiatives (leading to higher "E" scores) even if their absolute emissions remain substantial. (ii) Transparency versus Performance: Elevated "E" scores might primarily reflect enhanced transparency or reporting efforts rather than tangible emission reductions. (iii) Greenwashing: This finding could potentially signal "greenwashing," where companies effectively articulate firm environmental commitments in their public discourse without achieving proportional reductions in their actual operational emissions.

The analysis of Model (2) and particularly the environmental component in Model (3) does not confirm the hypothesis (H1) of a negative relationship between a firm's ESG discourse score (quantified by AI) and its real-world environmental performance (GHG emissions). Instead, for the aggregated ESG score and specifically the environmental component, we found a counterintuitive positive relationship. This suggests a discrepancy between a company's stated environmental discourse and its actual emissions, providing evidence that aligns with the phenomenon of greenwashing. Conversely, the social and governance components did show the expected negative relationship with emissions, indicating that good practices in these areas may genuinely contribute to lower environmental impact.

6 | Discussions of the Results

The analysis of the empirical results provides strong evidence of the practice of greenwashing in the Brazilian corporate sector, validating and expanding the theoretical framework on the dissonance between discourse and corporate practice. This discussion deepens the findings, using the theoretical framework of the "talk versus do" gap to interpret the results of the triangulation of data from:

6.1 | The Gap Between Discourse and External Perception: Analysis of Hypothesis 1

The econometric results revealed no significant correlation between AI-generated ESG scores and Merco ESG rankings (Model 1); failing to support Hypothesis 1. This absence of a statistical relationship indicates a substantial disconnect between corporate sustainability rhetoric and external market perception, strongly suggesting prevalent greenwashing practices. When examining specific cases, only 25% of firms with high ESG discourse scores (Table 5) maintained correspondingly strong Merco ESG reputations (Table 6), including notable examples like Itaú, Bradesco, and Ambev that demonstrated consistent alignment between communication and recognition. However, the majority (75%) of companies boasting extensive ESG terminology in their reports failed to achieve comparable Merco rankings, with Neonergia Group serving as a particularly striking example of this discourse-reputation gap. These findings collectively point to widespread discrepancies between sustainability claims and market validation across the Brazilian corporate sector.

6.2 | The Paradox Between Discourse and Practice: Analysis of Hypothesis 2

A more comprehensive analysis was conducted to investigate the "discourse versus practice" gap by comparing companies' ESG scores (Table 5) with their actual GHG emissions (Table 7). The evidence reveals a significant dissonance, as several companies that extensively discuss ESG practices in their reports are also among the most prominent GHG emitters. For instance,

15353966, 2025, 6, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/csr.70133, Wiley Online Library on [18/06/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 5 | The 20 companies with the highest occurrences of words in their ESG reports.

Companies	ESG score	Appear on the Merco ESG
Itau	0.284	X
Neonergia Group	0.284	
Cemig	0.281	
BB	0.279	X
Bradesco	0.273	X
Assai	0.263	
VALE	0.262	
Cpflra	0.261	
Raizen	0.252	
Copel	0.25	
Whirlpool	0.25	
Sabesp	0.248	
Petrobras	0.239	
Energy	0.237	
Sanepar	0.235	
Eletrobras	0.233	
Ambev	0.231	X
National Sid	0.229	
Gerdau	0.225	X
Telef Brazil	0.225	

Source: Based on ESG data by machine learning (2023).

TABLE 6 | The 20 companies with the highest ESG scores (according to ESG by Merco monitor).

Companies	Merco ESG
Natura Group	10,000
Itau	8017
Ambev	7980
Magalu	7771
Bradesco	7673
Itausa	6839
Renner Stores	6416
Safe Harbor	6393
Santanderbr	6293
Fleury	6222
BB	6049
BRF	5843
Suzanosa	5792
Gerdau	5656
BB Security	5571
Braskem	5416
Locate	5396
Cielo	5369
Santos BRP	5286
Raia drogasil	5268

Source: Based on the Merco ranking (2023).

Petrobras and Vale, which are listed among the companies with the highest frequency of ESG terms (Table 5), are also the top two largest emitters of greenhouse gases in the sample (Table 7). This contradiction is particularly notable given that both companies have previously been condemned for environmental and social damage. While it is crucial to consider that they operate in historically high-emitting sectors, this context raises further questions about their commitment to sustainability. This scenario exemplifies an apparent decoupling of discourse from practice and indicates that these companies must make greater efforts to reduce their emissions through cleaner technologies and more sustainable processes.

6.3 | Decoupling as the Underlying Mechanism of Greenwashing in Brazilian Companies

Our econometric analysis reveals significant decoupling between corporate ESG disclosures and GHG emissions, supporting greenwashing theories. This phenomenon, where organizations adopt formal policies without genuine implementation for legitimacy (Meyer and Rowan 1977), is demonstrated by triangulating AI-calculated ESG scores (internal discourse), Merco ESG index (external perception), and GHG emissions (actual practice). This aligns with greenwashing defined as a discrepancy between "responsible words" and "irresponsible actions" (Pizzetti et al. 2021), where firms mask performance (Lyon and Maxwell 2011) or exhibit extensive disclosure with poor actual results (Yu et al. 2020).

Model 2 shows companies with higher aggregate ESG scores tend to have higher GHG emissions, suggesting potential "greenwashing." Model 3 disaggregates ESG, paradoxically linking a higher Environmental (E) score to increased GHG emissions. Specifically, a 1% increase in a company's environmental discourse is associated with an approximate 5.24% increase in its GHG emissions. This "talking greener" while "being less green" corroborates that companies articulate environmental commitments without proportional reductions. This resonates with "carbonwashing" (In and Schumacher 2021) and observations that high ESG ratings don't necessarily mean lower carbon emissions (Treepongkaruna et al. 2024). It further validates concerns by Wu et al. (2020) and Parguel et al. (2011) that companies may strategically leverage the ESG concept to cultivate a false image of socio-environmental responsibility. Conversely, higher Social (S) and Governance (G) scores correlate with decreased GHG emissions, aligning with theoretical expectations. Strong social performance fosters operational efficiencies, while

15353966, 2025, 6, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/csr.70133, Wiley Online Library on [18/06/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 7 | The 20 companies with the highest greenhouse gas emissions per tCO_2e.

Companies	GHG
Petrobras	588,646,985.5
VALE	311,829,541.5
Gerdau	98,115,490.36
Vibra	96,056,073.26
JBS	61,144,421.33
CSN Mineracation	49,048,990.42
Braskem	28,747,796.74
National Sid	11,179,903.99
Usiminas	5,969,577.22
Eletrobras	5,640,034.53
Suzanosa	4,136,015.89
Simpar	3,779,157.68
Ambev	3,275,930.08
Eneva	2,696,279.34
Raizen	2,135,317.89
Let's Go	2,095,181.21
Neoenergia Group	1,788,457.94
Sabesp	1,654,721.93
Klabin Sa	1,515,080.14
Cosan	1,292,028.83

Source: Based on GHG emissions (2023).

robust governance enables better environmental risk management and emission reduction.

Corporate greenwashing suppresses employees' pro-environmental behavior. This occurs by inducing perceptions of corporate hypocrisy, negative pro-environmental attitudes, and organizational cynicism, leading to negative pro-environmental behaviors. These perceptions and attitudes are crucial mediators of greenwashing's influence on employee environmental engagement (Tao and Chao 2024). Keskin et al. (2025) demonstrate that employees with stronger green values are more sensitive to corporate greenwashing practices.

This study provides substantial evidence of greenwashing practices among Brazilian companies. Greenwashing, defined as false advertising creating a misleading sustainability image without concrete supportive actions (Seele and Gatti 2017; Ghitti et al. 2023), manifests prominently in our results. Some companies actively deceive stakeholders despite environmental concerns driving green product development (de Freitas Netto et al. 2020). Lyon and Maxwell (2011) describe "greenwash" as a firm strategically revealing positive environmental information while omitting negative aspects. The comparison of ESG performance with actual GHG emissions provides the most direct evidence of greenwashing. The presence of companies with robust ESG discourse and/or high external ESG ratings (e.g., Petrobras, Vale, Gerdau, Bradesco, Ambev) among the most significant GHG emitters directly illustrates this "dissonance between discourse and practice." This corroborates that companies claim environmental commitment while operating polluting facilities (Ghitti et al. 2023). For instance, Petrobras and Vale SA, despite frequent ESG term usage in reports, are major GHG emitters facing environmental legal action. This exemplifies how weak climate change regulation (Mateo-Márquez et al. 2022) can lead to opportunistic behavior, where companies maximize short-term profits without fully internalizing long-term environmental costs (Senna and de Araujo Moxotó 2025). This aligns with Wang et al. (2023)'s assertion that greenwashing harms sustainable development by making it challenging to identify genuinely sustainable companies.

In conclusion, the study's findings, particularly the positive elasticity of environmental discourse on emissions and the significant misalignment between internal ESG claims and actual environmental impact, empirically validate various facets of greenwashing theory. Consistent results across econometric models and comparative analyses, reinforced by corroborating studies using ML to detect greenwashing (Lien 2023; Lagasio 2023), underscore the urgent need for more robust regulation, standardized reporting, and independent verification to mitigate this deceptive practice and foster genuine corporate sustainability.

7 | Conclusion

Our study reveals substantial evidence of greenwashing among Brazilian B3-listed companies through decoupling, where formal sustainability policies exist without genuine implementation. This is consistently shown by triangulating internal discourse (AI-calculated ESG from corporate reports), external perception (Merco ESG index), and actual practice (GHG emissions). This aligns with greenwashing frameworks defining it as a disconnect between "responsible words" and "irresponsible actions" or strategic narrative management.

Our novel ML methodology, comparing AI-analyzed ESG indicators against external ratings and GHG emissions, yielded critical insights. B3-listed companies often disclose information inconsistent with actual environmental actions, strongly indicating greenwashing. OLS regression revealed no significant correlation between AI-calculated ESG scores from corporate discourse and external Merco ESG ratings, suggesting companies may exaggerate reported ESG practices. While a minority showed alignment, a substantial majority with high AI-ESG scores lacked similar external recognition, highlighting greenwashing's prevalence and the urgent need for transparent ESG practices.

Econometric models linking AI-derived ESG discourse to external GHG emission data showed paradoxical results. Higher overall ESG scores correlated with higher GHG emissions, suggesting image management may supersede actual environmental impact. A more detailed model reinforced this, showing increased AI-quantified environmental discourse correlated with increased emissions, indicating potential greenwashing or superficial transparency. In contrast, stronger social and

15353966, 2025, 6, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/csr.70133, Wiley Online Library on [18/06/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

governance discourse consistently linked to lower emissions, aligning with theoretical expectations. This illuminates a complex, often paradoxical, relationship between stated ESG commitments and actual environmental footprint, emphasizing the critical need for rigorous scrutiny.

This study's word frequency counting method has limitations in capturing full word context. However, its primary objective was proof of concept for a novel methodological framework to identify potential greenwashing by triangulating discourse, perception, and practice. The method's utility lies in revealing significant discrepancies, framing findings as "probability" of greenwashing rather than conclusive evidence. This validates our triangulation framework and sets an agenda for future research with more sophisticated NLP techniques.

Future research should enhance precision by including control variables (firm size, sector, financial performance) and larger samples. Developing a sophisticated, Portuguese-tailored corpus is crucial for analytical accuracy. High-quality databases are fundamental. Leveraging AI to combat greenwashing contributes to a more sustainable, transparent future.

8 | Policy Implications

The study highlights critical policy implications for greenwashing in Brazilian companies. Urgent needs include more rigorous regulatory frameworks due to current lax standards and unaudited data allowing misrepresentation. Policymakers should mandate independently audited carbon emissions disclosure for tangible reductions. Establishing clear, robust climate change regulations is vital for stakeholder confidence and accurate reporting, deterring short-term profit maximization over long-term environmental costs. Finally, policies should promote integrated ESG approaches, as strong social and governance practices correlate positively with emission reductions, fostering authentic sustainability beyond mere environmental discourse.

Acknowledgments

The authors would like to express their sincere gratitude to the Federal University of Amazonas (UFAM) for providing the necessary resources and an environment conducive to this research. We are deeply grateful to CAPES for the opportunity to publish our manuscript open access through the agreement with Wiley. The Article Processing Charge for the publication of this research was funded by the Coordenação de Aperfeiçoamento de Pessoal de Nível Superior - Brasil (CAPES) (ROR identifier: 00x0ma614).

References

Atilgan, Y., K. O. Demirtas, A. Edmans, and A. D. Gunaydin. 2024. "Does the Carbon Premium Reflect Risk or Mispricing?" FEB-RN Research Paper, (03).

Baldi, F., and A. Pandimiglio. 2022. "The Role of ESG Scoring and Greenwashing Risk in Explaining the Yields of Green Bonds: A Conceptual Framework and an Econometric Analysis." *Global Finance Journal* 52: 100711.

Bothello, J., I. Ioannou, V. A. Porumb, and Y. Zengin-Karaibrahimoglu. 2023. "CSR Decoupling Within Business Groups and the Risk of Perceived Greenwashing." *Strategic Management Journal* 44, no. 13: 3217–3251.

Boykoff, J., and G. Mascarenhas. 2016. "The Olympics, Sustainability, and Greenwashing: The Rio 2016 Summer Games." *Capitalism Nature Socialism* 27, no. 2: 1–11.

Braga Junior, S., M. P. Martínez, C. M. Correa, R. C. Moura-Leite, and D. Da Silva. 2019. "Greenwashing Effect, Attitudes, and Beliefs in Green Consumption." *RAUSP Management Journal* 54, no. 2: 226–241.

Brito, A. C. F. D. M., S. L. F. G. Dias, and E. S. Zaro. 2022. "Corporate Socioenvironmental Report and Greenwashing: Analysis of a Brazilian Mining Company." *Cadernos Ebape. Br* 20: 234–246.

Bromley, P., and W. W. Powell. 2012. "From Smoke and Mirrors to Walking the Talk: Decoupling in the Contemporary World." *Academy of Management Annals* 6, no. 1: 483–530.

Caldas, M. V. D. A., A. R. Veiga-Neto, L. G. D. A. Guimarães, A. B. C. D. Castro, and G. R. B. Pereira. 2021. "Greenwashing in Environmental Marketing Strategy in the Brazilian Furniture Market." *Revista de Economia e Sociologia Rural* 59, no. 3: e225336.

Chen, P., and A. A. Dagestani. 2023. "Greenwashing Behaviour and Firm Value–From the Perspective of Board Characteristics." *Corporate Social Responsibility and Environmental Management* 30, no. 5: 2330–2343.

Das, P., V. V. Themmadath, S. Jeyalakshmi, K. Lateef, and A. Agnihotri. 2025. "AI in Sustainable Finance: Identifying Green Investments Through Machine Learning." *Advances in Consumer Research* 2, no. 3: 80.

de Freitas Netto, S. V., M. F. F. Sobral, A. R. B. Ribeiro, and G. R. D. L. Soares. 2020. "Concepts and Forms of Greenwashing: A Systematic Review." *Environmental Sciences Europe* 32, no. 1: 19.

Delmas, M. A., and V. C. Burbano. 2011. "The Drivers of Greenwashing." *California Management Review* 54, no. 1: 64–87.

Duarte, J. P., S. A. Pereira Bastos, M. J. d'Angelo, E. S. de Oliveira, and G. Fortunato. 2024. "Oversight Mechanisms and the Reduction of Greenwashing." *Environmental & Social Management Journal/Revista de Gestão Social e Ambiental* 18, no. 8: e06098.

Gao, J., and H. Wei. 2025. "Study on the Impact of Environmental Subsidies and Green Labels on Greenwashing in Sustainable Production." *Journal of Cleaner Production* 486: 144531.

Ghitti, M., G. Gianfrate, and L. Palma. 2023. "The Agency of Greenwashing." *Journal of Management and Governance* 28, no. 3: 1–941.

In, S. Y., and K. Schumacher. 2021. "Carbonwashing: ESG Data Greenwashing in a Post-Paris World." In *Settling Climate Accounts: Navigating the Road to Net Zero*, 39–58. Springer International Publishing.

Kalesnik, V., M. Wilkens, and J. Zink. 2020. "Green Data or Greenwashing? Do Corporate Carbon Emissions Data Enable Investors to Mitigate Climate Change?" SSRN Electronic Journal.

Kathan, M. C., S. Utz, G. Dorfleitner, J. Eckberg, and L. Chmel. 2025. "What You See Is Not What You Get: ESG Scores and Greenwashing Risk." *Finance Research Letters* 74: 106710.

Keresztúri, J. L., E. Berlinger, and Á. Lublóy. 2025. "Environmental Policy and Stakeholder Engagement: Incident-Based, Cross-Country Analysis of Firm-Level Greenwashing Practices." *Corporate Social Responsibility and Environmental Management* 32, no. 1: 192–211.

Keskin, H., E. Tatoglu, A. E. Akgün, E. Esen, and E. Salt. 2025. "Employee Green Values and Perceived Greenwashing: The Mediating Role of Person-Organization Fit in CSR-Oriented Companies." *Corporate Social Responsibility and Environmental Management* 32, no. 4: 4861–4880.

Khan, M. K., C. Huo, R. A. Zahid, and U. S. Maqsood. 2024. "The Automated Sustainability Auditor: Does Artificial Intelligence Curtail

15353966, 2025, 6, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/csr.70133, Wiley Online Library on [18/06/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Greenwashing Behaviour in Chinese Firms?" *Business Strategy and the Environment* 33, no. 8: 9015–9039.

Lagasio, V. 2023. "Measuring Greenwashing: The Greenwashing Severity Index." Available at SSRN 4582917.

Li, D., Z. Zhang, and X. Gao. 2024. "Does Artificial Intelligence Deter Greenwashing?" *Finance Research Letters* 67: 105954.

Lien, A. S. 2023. "Machine Learning to Detect Corporate Greenwashing." (Master's Thesis, UIS).

Liu, G., H. Qian, Y. Shi, D. Yuan, and M. Zhou. 2024. "How Do Firms React to Capital Market Liberalization? Evidence From ESG Reporting Greenwashing." *Corporate Social Responsibility and Environmental Management* 31, no. 5: 4329–4344.

Luu, N. H., C. Le, H. N. Luu, and D. T. Nguyen. 2025. "Does Mandatory Greenhouse Gas Emissions Reporting Program Deter Corporate Greenwashing?" *Journal of Environmental Management* 373: 123740.

Lyon, T. P., and J. W. Maxwell. 2011. "Greenwash: Corporate Environmental Disclosure Under Threat of Audit." *Journal of Economics and Management Strategy* 20, no. 1: 3–41.

Mansouri, S., and P. P. Momtaz. 2022. "Financing Sustainable Entrepreneurship: ESG Measurement, Valuation, and Performance." *Journal of Business Venturing* 37, no. 6: 106258.

Mateo-Márquez, A. J., J. M. González-González, and C. Zamora-Ramírez. 2022. "An International Empirical Study of Greenwashing and Voluntary Carbon Disclosure." *Journal of Cleaner Production* 363: 132567.

Medeiros, P., D. Salles, T. Magalhães, B. Melo, and R. M. Santini. 2024. "Greenwashing and Disinformation: The Toxic Advertising of Brazilian Agribusiness on Social Media." *Comunicação e Sociedade* 45: 1–26.

Meyer, J. W., and B. Rowan. 1977. "Institutionalised Organisations: Formal Structure as Myth and Ceremony." *American Journal of Sociology* 83, no. 2: 340–363.

Mishra, P., and G. B. Schmidt. 2018. "How Can Leaders of Multinational Organisations Be Ethical by Contributing to Corporate Social Responsibility Initiatives? Guidelines and Pitfalls for Leaders Trying to Do Well." *Business Horizons* 61, no. 6: 833–843.

Moodaley, W., and A. Telukdarie. 2023. "Greenwashing, Sustainability Reporting, and Artificial Intelligence: A Systematic Literature Review." *Sustainability* 15, no. 2: 1481.

Mougenot, B., and J. P. Doussoulin. 2022. "The Conceptual Evolution of the Economy: A Bibliometric Analysis." *Environment, Development and Sustainability* 24, no. 1: 1031–1047.

Paprocka, M. W. 2025. "Navigating Ethical Dilemmas: Unveiling Greenwashing in the AI Era." In *Building Trust in the Generative Artificial Intelligence Era*, 34–42. Routledge.

Parguel, B., F. Benoît-Moreau, and F. Larceneux. 2011. "How Sustainability Ratings Might Deter 'Greenwashing': A Closer Look at Ethical Corporate Communication." *Journal of Business Ethics* 102: 15–28.

Pizzetti, M., L. Gatti, and P. Seele. 2021. "Firms Talk, Suppliers Walk: Analysing the Locus of Greenwashing in the Blame Game and Introducing 'vicarious Greenwashing'." *Journal of Business Ethics* 170, no. 1: 21–38.

Seele, P., and L. Gatti. 2017. "Greenwashing Revisited: In Search of a Typology and Accusation-Based Definition Incorporating Legitimacy Strategies." *Business Strategy and the Environment* 26, no. 2: 239–252.

Senna, A. L., and A. C. de Araujo Moxotó. 2025. "Carbon Emissions and Financial Performance in the Brazilian Stock Market." *Journal of Environmental Management* 377: 124698.

Szabo, S., and J. Webster. 2021. "Perceived Greenwashing: The Effects of Green Marketing on Environmental and Product Perceptions." *Journal of Business Ethics* 171: 719–739.

Tao, Z., and J. Chao. 2024. "Unveiling the Influence of Corporate Greenwashing on Employees' Pro-Environmental Behavior: A Cross-Cultural Study From China, United Kingdom, Republic of Korea, and Japan." *Corporate Social Responsibility and Environmental Management* 31, no. 6: 6103–6130.

Toscano, A., M. Balzarotti, and I. Re. 2022. "Sustainability Practices and Greenwashing Risk in the Italian Poultry Sector: A Grounded Theory Study." *Sustainability* 14, no. 21: 14088.

Treepongkaruna, S., H. H. Au Yong, S. Thomsen, and K. Kyaw. 2024. "Greenwashing, Carbon Emission, and ESG." *Business Strategy and the Environment* 33, no. 8: 8526–8539.

Wang, W., Z. Sun, W. Zhu, et al. 2023. "How Does Multi-Agent Governance Govern Corporate Greenwashing? A Stakeholder Engagement Perspective From "Common" to "Collaborative" Governance." *Corporate Social Responsibility and Environmental Management* 30, no. 1: 291–307.

Wu, Y., K. Zhang, and J. Xie. 2020. "Bad Greenwashing, Good Greenwashing: Corporate Social Responsibility and Information Transparency." *Management Science* 66, no. 7: 3095–3112.

Yu, E. P. Y., B. Van Luu, and C. H. Chen. 2020. "Greenwashing in Environmental, Social, and Governance Disclosures." *Research in International Business and Finance* 52: 101192.

Zeng, F., J. Wang, and C. Zeng. 2025. "An Optimised Machine Learning Framework for Predicting and Interpreting Corporate ESG Greenwashing Behaviour." *PLoS One* 20, no. 3: e0316287.

Zhang, D. 2024. "The Pathway to Curb Greenwashing in Sustainable Growth: The Role of Artificial Intelligence." *Energy Economics* 133: 107562.

15353966, 2025, 6, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/csr.70133, Wiley Online Library on [18/06/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License